# Annex D: DDIL & Classification Reference

Challenge 2: The Quantum Routing Brain

Tactical network constraints, data classification levels, and cross-domain considerations that govern when and how your routing engine can offload compute to a remote quantum backend.

## DDIL: THE NETWORK REALITY

DDIL stands for Disconnected, Degraded, Intermittent, and Limited. It describes the network conditions that tactical military systems operate under routinely, not as edge cases. Your routing engine must treat DDIL as the default operating condition, not the exception.

**Why this matters for quantum offload.** Every quantum backend your router could dispatch to (IBM Quantum, AWS Braket, Azure Quantum, D-Wave Leap) is hosted in a commercial data center in CONUS, Europe, or Asia. Reaching that data center from a forward-deployed tactical node requires traversing one or more of the links below. If the link drops mid-computation, your router must handle the failure gracefully: fall back to local classical, retry when connectivity restores, and never lose the task.

### Tactical Link Budget Reference

| Link Type | RTT Latency | Bandwidth | Characteristics |
|---|---|---|---|
| GEO MILSATCOM (WGS, AEHF) | 500–600 ms | 1–10 Mbps tactical | High latency, moderate bandwidth. The workhorse for beyond-line-of-sight. Vulnerable to jamming and weather. |
| MEO (O3b mPOWER) | 50–150 ms | Multi-Gbps (shared) | Lower latency than GEO. Limited military adoption. Commercial service. |
| LEO (Starlink / Starshield) | 20–50 ms | 100–150 Mbps | Lowest SATCOM latency. Starshield is the DoD variant. Constellation availability varies by theater. |
| MUOS (UHF SATCOM) | GEO-class | Up to 384 kbps/user | Narrowband. Voice and low-rate data. Not suitable for quantum job payloads except very small QUBO encodings. |
| Link 16 / TADIL-J | LOS, slot-bounded | 31.6 / 57.6 / 115.2 kbps | Time-division data link. 1,536 slots per 12-second epoch. Message-oriented, not stream-oriented. Too constrained for quantum offload. |
| Tactical mesh (Silvus StreamCaster, Persistent Wave Relay MPU5) | <45 ms node-to-node | Up to ∼100 Mbps TCP | MN-MIMO mesh. Good within a tactical enclave. Does not reach CONUS. |
| Tactical 5G (fixed base) | <10–15 ms | 1.5–3.9 Gbps demonstrated | Low latency, high bandwidth. Only available at fixed installations with 5G infrastructure. DoD smart warehouse demos. |
| HF wideband (MIL-STD-188-110D) | Seconds (BLOS) | Up to 120 kbps over 24 kHz | Beyond-line-of-sight without satellite. Very low bandwidth. Backup/last-resort link. |

### Implications for Your Router

> **Rule of thumb:** If the task's latency budget is under 500 ms, it cannot tolerate a GEO SATCOM round trip. If under 50 ms, it cannot tolerate any SATCOM. If under 10 ms, it must run on the local edge device. Only batch tasks (seconds to hours) can realistically reach a CONUS quantum cloud and return results.

Your router should implement at minimum:

- **Link health check.** Before every offload decision, ping or estimate round-trip time to the quantum backend. If the link is down or degraded beyond the task's latency budget, route classical immediately.
- **Graceful degradation.** If the link drops after a job is submitted but before results return, the router must not block. Queue the result for later retrieval and run the classical fallback now.
- **Bandwidth awareness.** A QUBO encoding for a 50-variable problem is a few KB. A full I/Q dataset for quantum kernel classification could be MB. Over a 384 kbps MUOS link, the I/Q transfer alone takes minutes. Factor payload size into the routing decision.
- **Retry with backoff.** Intermittent links come back. Your router should retry failed offloads with exponential backoff, not abandon them permanently.

### Simulating DDIL in Your Demo

You do not need real satellite links. Simulate DDIL conditions in your Docker Compose environment:

- **Latency injection:** Use `tc netem` (Linux traffic control) to add 250–600 ms delay on the network interface between your router container and the quantum backend container. `tc qdisc add dev eth0 root netem delay 500ms 100ms distribution normal`
- **Bandwidth throttling:** Limit bandwidth to simulate SATCOM. `tc qdisc add dev eth0 root tbf rate 1mbit burst 10kb latency 500ms`
- **Link drops:** Use `tc netem` packet loss or simply `docker stop` the quantum backend container mid-demo to show graceful fallback. `tc qdisc add dev eth0 root netem loss 30%`
- **Intermittent connectivity:** Script periodic link drops and restorations to show retry behavior.

The compelling demo moment: tasks are flowing, the link to the quantum backend drops, the router immediately falls back to classical for all pending tasks, results continue to flow, and when the link restores, the router resumes quantum offload for eligible tasks. That is operational resilience.

## DATA CLASSIFICATION LEVELS

The U.S. Department of Defense classifies information into levels that directly constrain where data can be processed. The DISA Cloud Computing Security Requirements Guide (CC SRG) maps these classification levels to Impact Levels (IL) that determine which cloud environments are authorized.

### Impact Level Definitions

| IL | Data Type | Authorized Environments | Quantum Offload? |
|---|---|---|---|
| **IL2** | Public, non-CUI | Any FedRAMP Moderate cloud (AWS GovCloud, Azure Gov, Google Cloud) | Yes. Any commercial QPU. |
| **IL4** | CUI (Controlled Unclassified Information) | FedRAMP High + DoD CC SRG IL4 authorization | Conditional. Must verify QPU provider's IL4 authorization. |
| **IL5** | Higher-sensitivity CUI and unclassified National Security Systems (U-NSS) | IL5-authorized clouds. Google Cloud was first hyperscaler authorized (2022). AWS GovCloud and Azure Gov also hold IL5. | Conditional. This is the maximum level at which commercial quantum services could theoretically operate. No QPU provider has published IL5 authorization as of April 2026. |
| **IL6** | SECRET / NSS. SIPRNet-resident. | On-premises or IL6-authorized cloud (AWS Secret Region, Azure Government Secret, Google Distributed Cloud) | **No.** No commercial QPU is authorized at IL6. SECRET data cannot leave the classified enclave for quantum processing. |
| **–** | TS-SCI | SCIFs, JWICS-connected systems, special-access enclaves | **No.** Not offloadable under any current or foreseeable commercial accreditation. |

### The IL5 Ceiling

**The structural constraint.** Commercial quantum services (IBM Quantum, AWS Braket, Azure Quantum, D-Wave Leap) operate their QPU hardware in standard commercial data centers. None of these facilities hold IL6 authorization. Even IL5 authorization for QPU-specific services has not been publicly confirmed by any provider as of April 2026. Your router must treat IL5 as the absolute ceiling for quantum offload and should conservatively default to IL4 (standard CUI) unless you can verify the specific provider's authorization level.
In practice, this means: any task involving SECRET or TS-SCI data is not offloadable to a commercial QPU under any circumstances. Period. Your router must enforce this as a hard gate, not a preference.

### Cross-Domain Solutions

When a classified workflow contains a quantum-eligible sub-problem, the only path to offload is decomposition: extract the sub-problem, strip all classification-sensitive context, verify the sanitized payload is releasable at IL4/IL5, offload the sanitized problem to the QPU, and reintegrate the result into the classified workflow.

This is the pattern your Data Guard (stretch goal) implements. Key concepts:

**Cross-Domain Solution (CDS)** — A system that enables information transfer between security domains (e.g., SECRET to UNCLASS) with enforced content filtering, audit, and policy controls. Governed by NSA's National Cross Domain Strategy and Management Office (NCDSMO).

**Raise the Bar (RTB)** — NSA's security baseline for cross-domain solutions. Requires Redundant, Always-invoked, Independent implementations, Non-bypassable (RAIN principles). Mandated by NSM-8 (January 2022) for any CDS connected to a National Security System.

**Guard** — A CDS component that inspects data crossing a security boundary and enforces content-based filtering rules. In your demo, the Data Guard stub plays this role: it inspects the outbound quantum job request, redacts classification-sensitive fields, and allows only the sanitized mathematical problem (e.g., a QUBO adjacency matrix with no operational labels) to cross to the commercial QPU.

> **Scope for the demo.** You are not building a real CDS. Real cross-domain solutions require NSA certification and years of development. You are building a *conceptual stub* that demonstrates the decomposition pattern: show that your router can take a SECRET-labeled task, extract the quantum-eligible sub-problem as an unlabeled mathematical object, flag it as UNCLASS, route it to the QPU, and reattach the operational context when the result returns. The pattern is the deliverable, not the certification.

## CURSOR ON TARGET (COT) MESSAGE FORMAT

Cursor on Target is the standard open tactical message format used across the TAK ecosystem (ATAK, WinTAK, iTAK, FreeTAKServer). If your demo uses a tactical data stream, CoT is the natural message envelope.

### Overview

**Origin** — Developed by MITRE (2003). Public XSD schema available at `https://github.com/docjason/XmlValidate` (Event.xsd v2.0). XML-based.

**Message size** — A basic Position Location Information (PLI) event is 300–800 bytes. Rich detail extensions (sensor SPI, chat, MEDEVAC, C-UAS tracks) reach 1,000–2,000 bytes.

**Update rate** — ATAK clients typically send position updates at 1 Hz. Sensor and track updates vary by source.

**TAK ecosystem** — ATAK (Android), WinTAK (Windows), iTAK (iOS), TAK Server (Java, USG open-sourced), FreeTAKServer (Python open-source), taky (lightweight Python server). All freely available.

### Structure

A CoT event has three core elements:

- `<event>`: root element with `uid` (unique identifier), `type` (CoT type string, e.g., `a-f-G-U-C` for a friendly ground unit), `time`, `start`, `stale` (validity window), and `how` (how the position was determined).
- `<point>`: latitude, longitude, altitude (HAE), circular error (CE), and linear error (LE).
- `<detail>`: extensible element for mission-specific data. This is where sensor readings, track metadata, classification labels, and quantum routing metadata would go.

### Using CoT in Your Demo

Wrap each task in your synthetic data stream as a CoT event. The `<detail>` element carries the task metadata from the CSV (task type, classification level, latency budget, quantum candidacy). Your router parses the CoT event, extracts the routing-relevant fields, makes the dispatch decision, and logs the result as a new CoT event with the routing decision in the `<detail>` block.

This is not required. Your demo can use plain JSON task envelopes and be perfectly valid. But CoT formatting adds tactical realism that scores points on the Defense Realism criterion and demonstrates awareness of how real tactical systems exchange data.

## JADC2 AND TACTICAL COMPUTE ARCHITECTURE

For teams that want to ground their architecture in the broader DoD compute context.

**JADC2 (Joint All-Domain Command and Control)** — DoD's concept for connecting sensors, shooters, and decision-makers across all domains (air, land, sea, space, cyber) into a unified command and control fabric. Public strategy summary: `https://media.defense.gov/2022/Mar/17/2002958406/-1/-1/1/SUMMARY-OF-THE-JOINT-ALL-DOMAIN-COMMAND-AND-CONTROL-STRATEGY.pdf`. The implementation plan is classified.

**ABMS (Advanced Battle Management System)** — The Air Force's contribution to JADC2. GAO-23-105495 and DOT&E FY2024 reports document status. CBC2 (homeland air defense for NORAD/NORTHCOM) is the only ABMS component with scheduled operational testing.

**CJADC2 Minimum Viable Capability** — Announced by Deputy Secretary Hicks on 21 February 2024. GAO-25-106454 (2025) found DoD has yet to establish a guiding framework. The architecture is still being defined, which is both the challenge and the opportunity for any company building integration infrastructure.

**Federated Data Fabric** — JADC2's working concept for data sharing across domains, echelons, and security levels. No single authoritative data store. Instead, data is discoverable and accessible across federated nodes. Your quantum routing engine fits naturally into this architecture as a compute service discoverable by any node in the fabric.

**NATO Federated Mission Networking (FMN)** — NATO's framework for interoperable coalition networking. Spiral 5 Standards Profile defines message formats, security profiles, and interoperability layers. Public PDF at `https://storage.nisp.nw3.dk/20221202_FMN_Spiral_5_Standards_Profile.pdf`. Relevant for Thread 3 (Coalition PQC Migration).

**Mission Partner Environment (MPE)** — DoD's framework for sharing data with coalition partners at varying classification levels. DoDI 8110.01 governs. MPE-S consolidates legacy coalition networks (CENTRIXS, CFBLNet, BICES, Pegasus, APAN) connecting approximately 45,000 users globally.

## DOD ZERO TRUST ARCHITECTURE

If your demo implements any identity or access control for the quantum backends, align it with DoD Zero Trust concepts.

**DoD Zero Trust Strategy** — Published 21 October 2022 at `https://dodcio.defense.gov/Portals/0/Documents/Library/`. Seven pillars: User, Device, Application/Workload, Data, Network/Environment, Automation/Orchestration, Visibility/Analytics.

**Zero Trust Reference Architecture v2.0** — Published July/September 2022. Same URL. Defines the target architecture for DoD network modernization.

**NIST SP 800-207** — Zero Trust Architecture (August 2020). The foundational federal reference. `https://csrc.nist.gov/publications/detail/sp/800-207/final`.

**SPIFFE/SPIRE** — CNCF Graduated project for workload identity. Issues SVIDs (SPIFFE Verifiable Identity Documents) to services without relying on network location. Integrates with Istio for mTLS. Your router could use SPIFFE identities to verify that the quantum backend container is who it claims to be before sending task data. `https://spiffe.io/`.

> **For your demo:** You do not need to implement a full zero-trust architecture. But if your router checks that the quantum backend container presents a valid identity before accepting offload requests, and if your audit log records which identity processed each task, that demonstrates awareness of how real defense systems handle trust. These are small additions with high scoring impact on Defense Realism.